

Ethanol Blending in Gasoline - Mexico

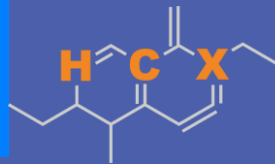
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



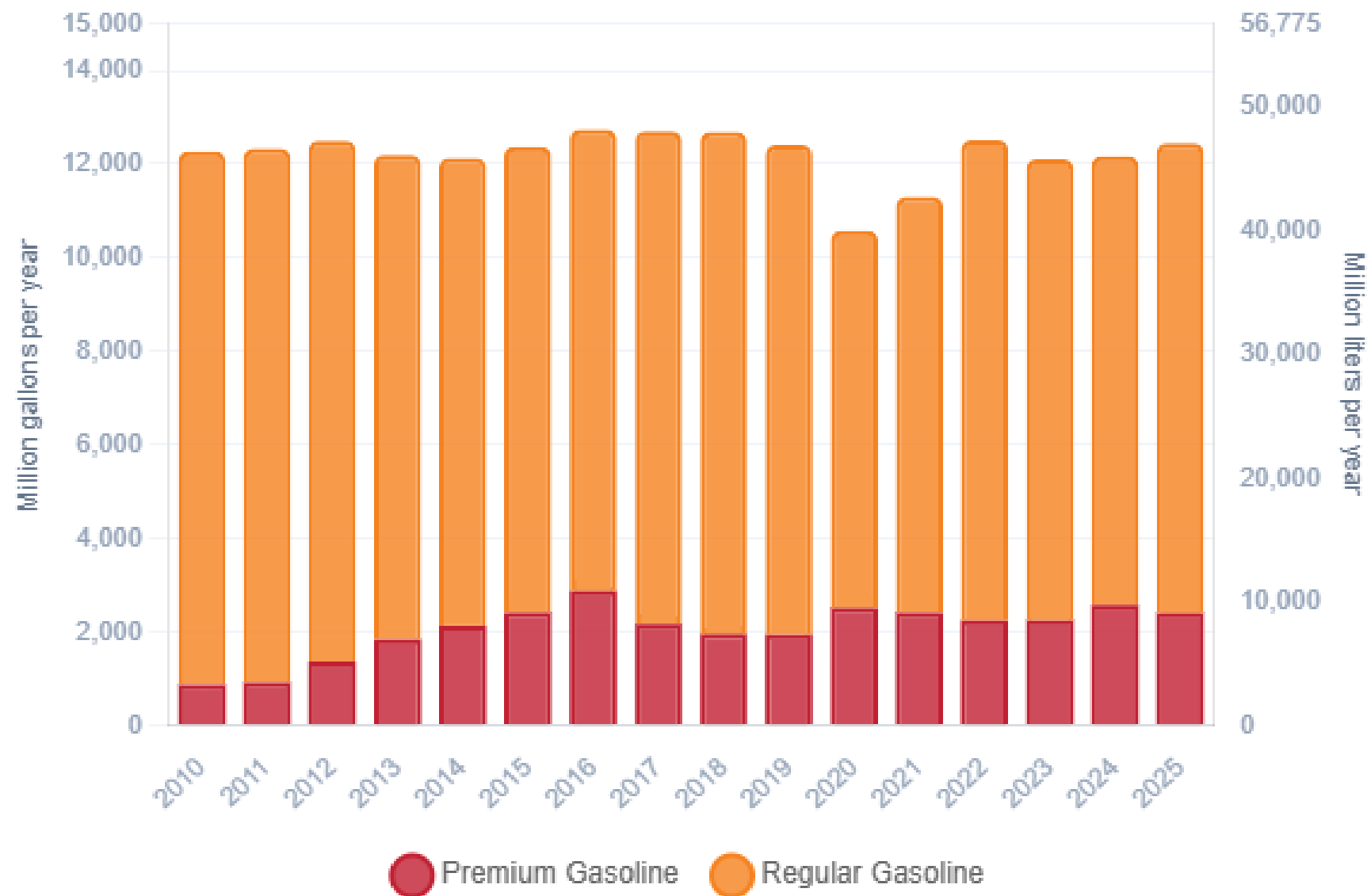
Ethanol Blending in Gasoline - Mexico



In 2025, gasoline consumption was 12,447 million gallons (47,116 million liters). There are two grades of gasoline, Regular (AKI 87) and Premium (AKI 91). Regular gasoline is the main grade in the market, representing 80% of the volume. There are stringent gasoline specifications for major metropolitan areas in Mexico City, Guadalajara and Monterrey. Demand is much higher than production, making necessary to import 40% of gasoline mainly from the United States.

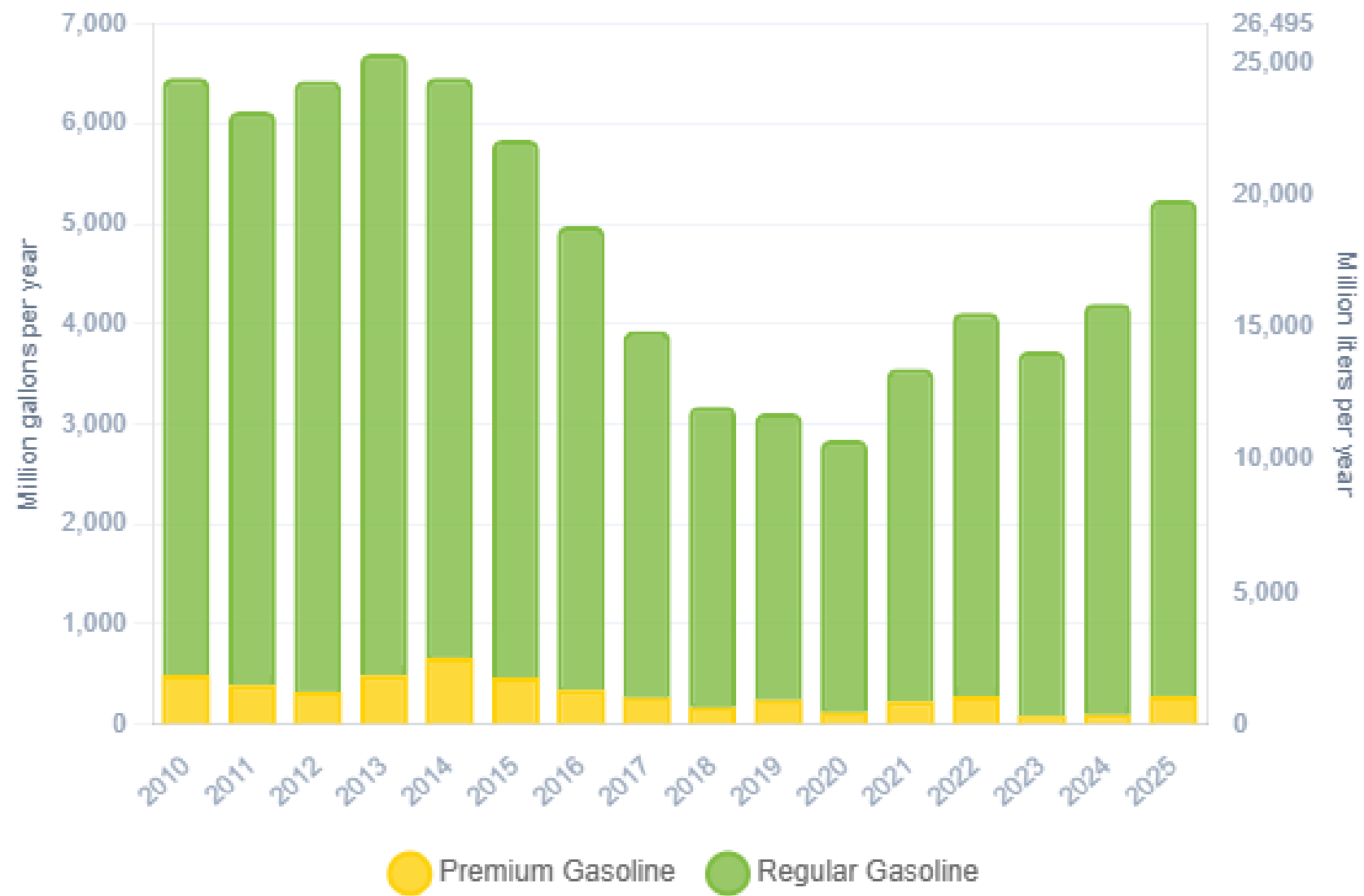
Gasoline quality norm (NOM-016-CRE de 2016) allows blends with a maximum of 5.8% v/v of ethanol with the exception of metropolitan areas of Valle de Mexico, Guadalajara and Monterrey. Current ethanol use is equivalent to an average of 1.06% in the Rest of the Country grade gasoline. In 2025, a new biofuels law was approved that fosters the production with agricultural and other organic residuals. Recently there are new state level efforts to increase ethanol consumption (e.g. Tamaulipas).

Gasoline Demand in Mexico



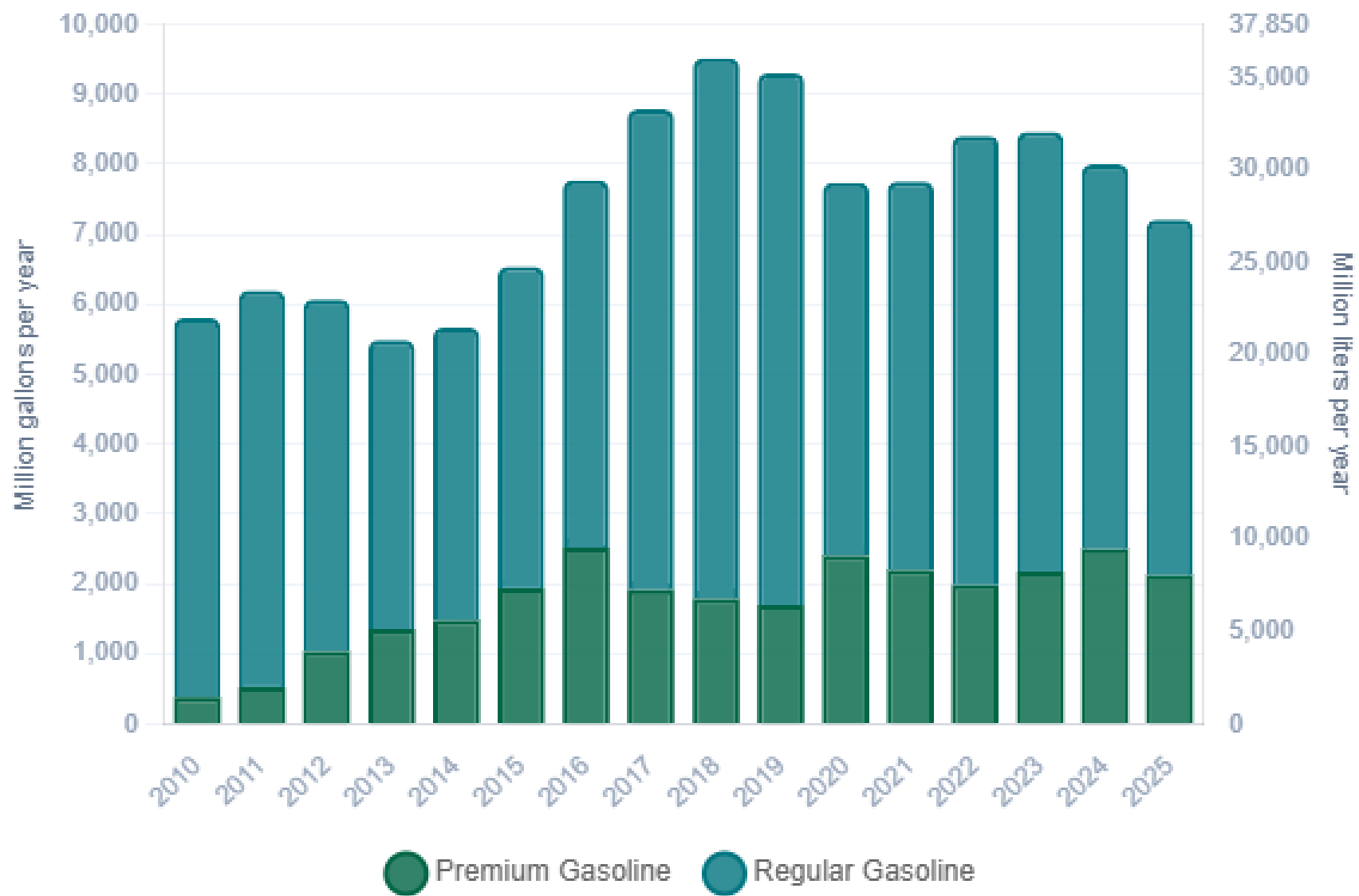
Source: SIE - SENER, 2025

Gasoline Production in Mexico



Source: SE - SENER, 2025

Gasoline Imports in Mexico



Source: SE - SENER, 2025

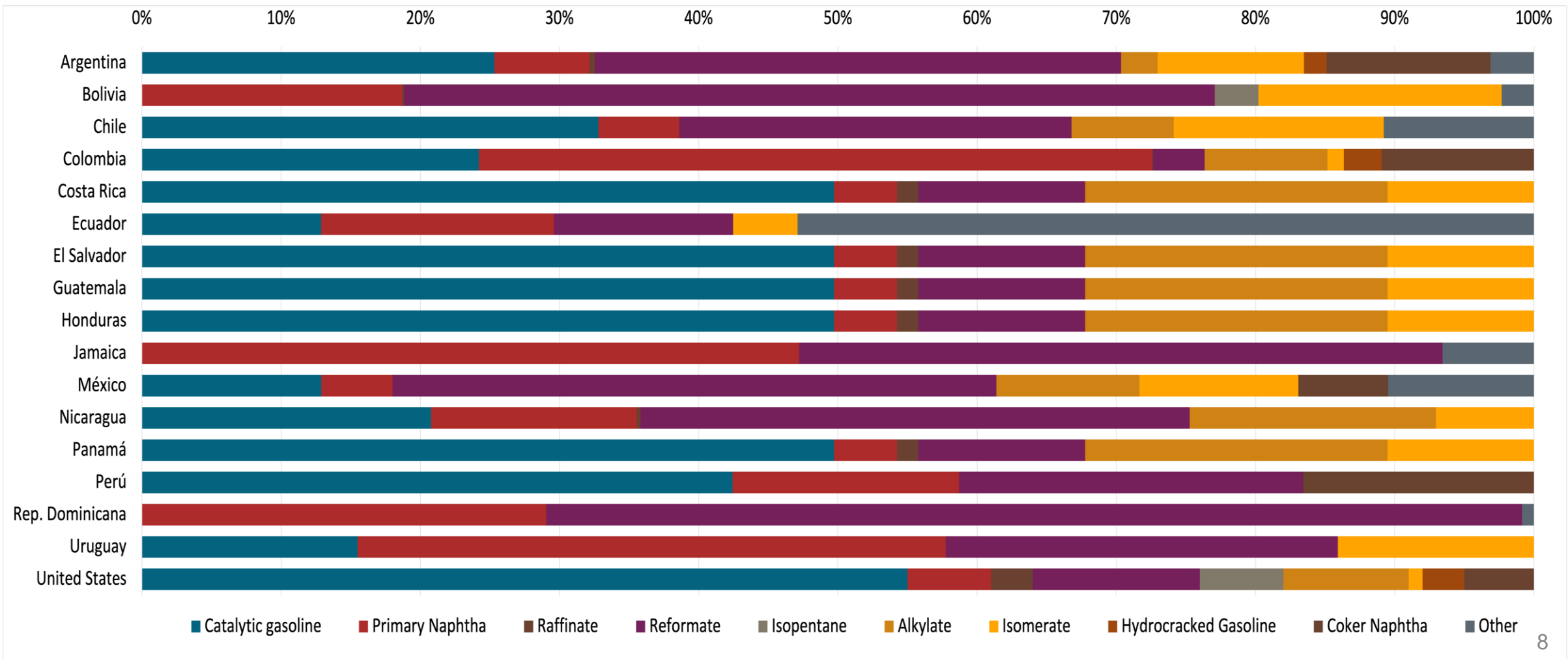
Gasoline Quality in Mexico

Name	NOM-016-CRE-2016						EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2016						2017			
Applicability	Mexico City	Mexico City	Guadalajara and Monterrey	Guadalajara and Monterrey	Rest of the country	Rest of the country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	Gasoline Regular	Gasoline Premium	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 2 %v/v	< 2 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 25 %v/v	< 25 %v/v	< 32 %v/v	22,6% v/v	Report	22,6% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 10 %v/v	< 10 %v/v	< 11,9 %v/v	< 11,9 %v/v	Report	< 2,5 mg/l	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	-	-	-	-	-	-	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0 mg/l	0 mg/l	0 mg/l	0 mg/l	0 mg/l	0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON		94		94		94	> 95	> 95	> 98	> 98
MON	82		82		82		> 85	> 88	> 85	> 88
AKI										
Sulfur Content	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m
Ethanol (EtOH)	Not allowed	Not allowed	Not allowed	Not allowed	> 5,8 %v/v	> 5,8 %v/v	< 5 %v/v	< 10 %v/v	< 5 %v/v	< 10 %v/v
RVP 37.8°C (Summer)	< 62 kPa	< 62 kPa	< 62 kPa	< 62 kPa	< 69 kPa North, < 62 kPa Center, South, Pacific	< 69 kPa North, < 62 kPa Center, South, Pacific	< 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa				
RVP 37.8°C (Transition)	< 54 kPa	< 54 kPa	< 54 kPa Guadalajara	< 54 kPa Guadalajara						
MTBE							-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	-	-	Based on oxygen content	< 22 %v/v	Based on oxygen content	< 22 %v/v

Source: CRE, 2016

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



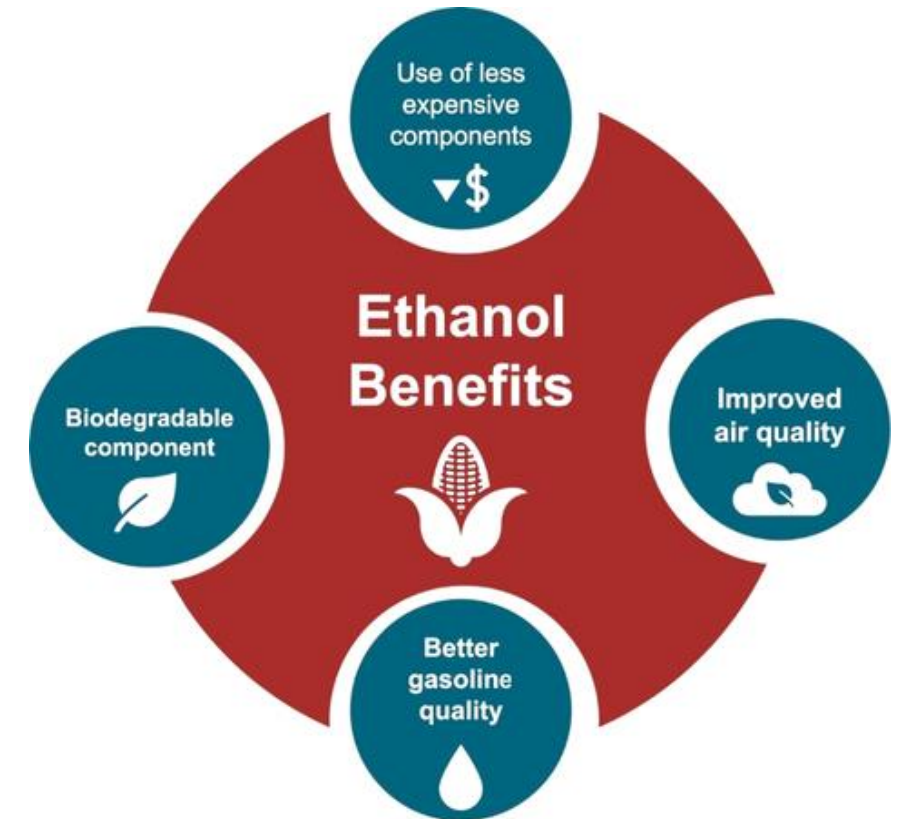
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

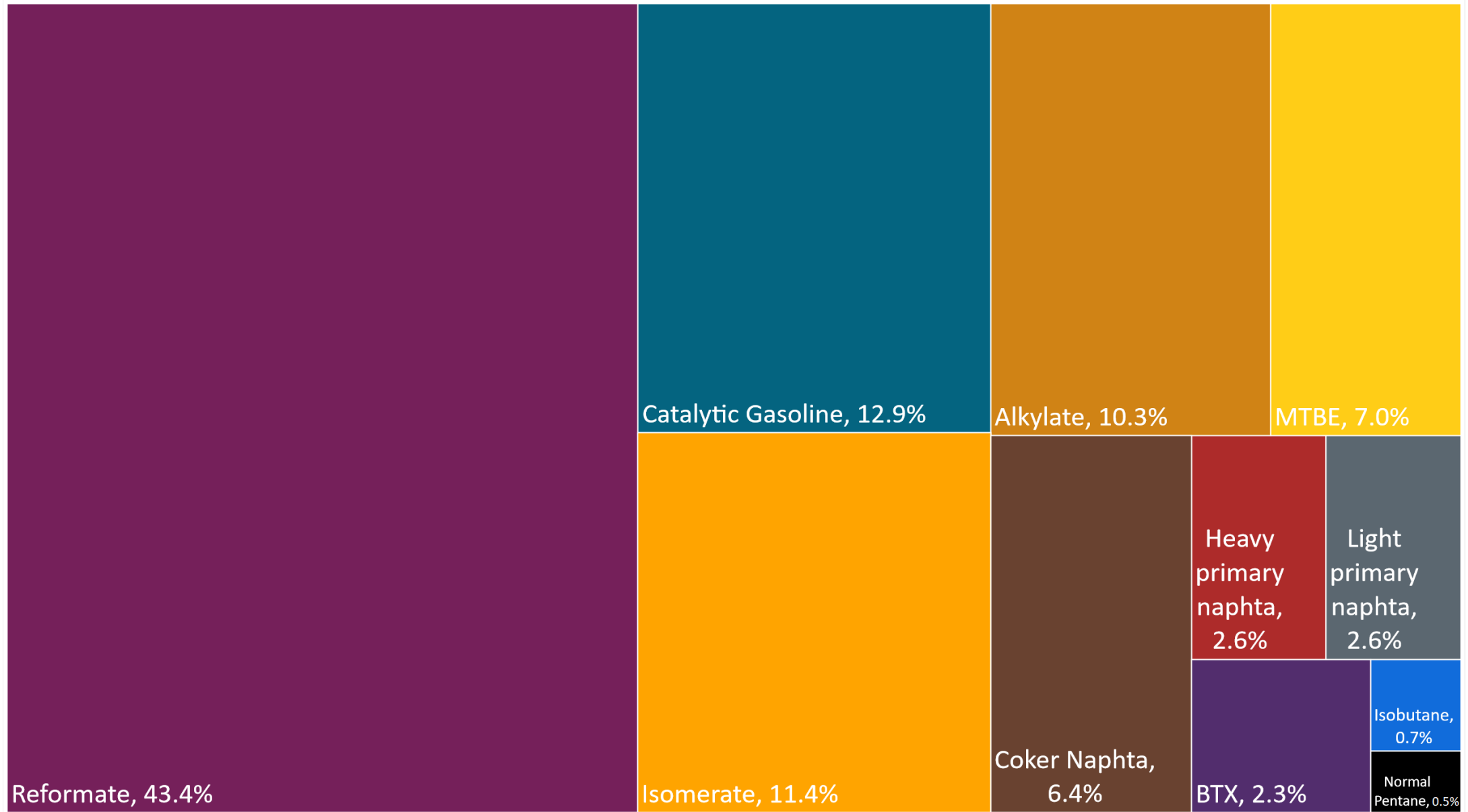
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

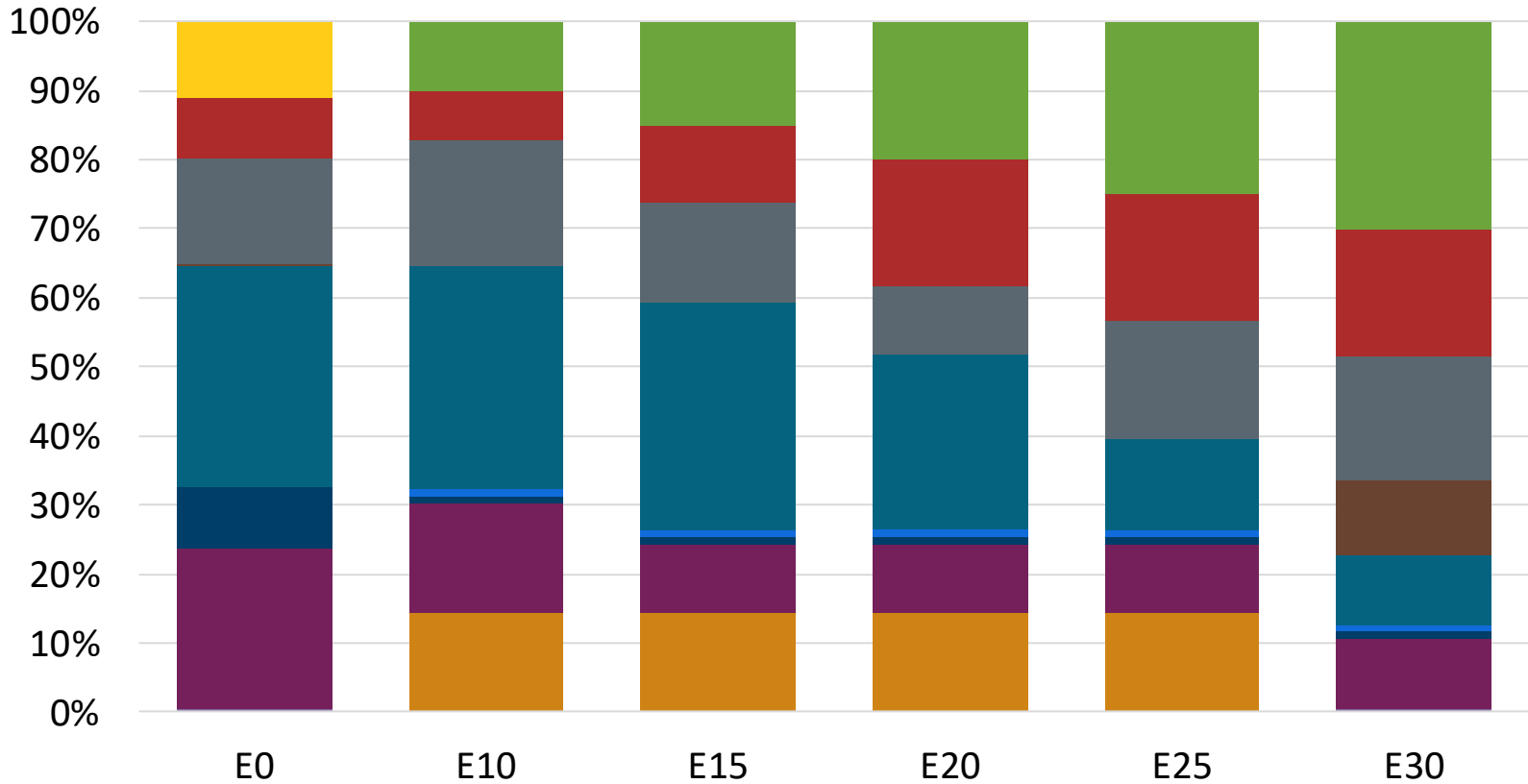
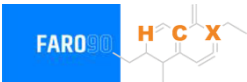
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components

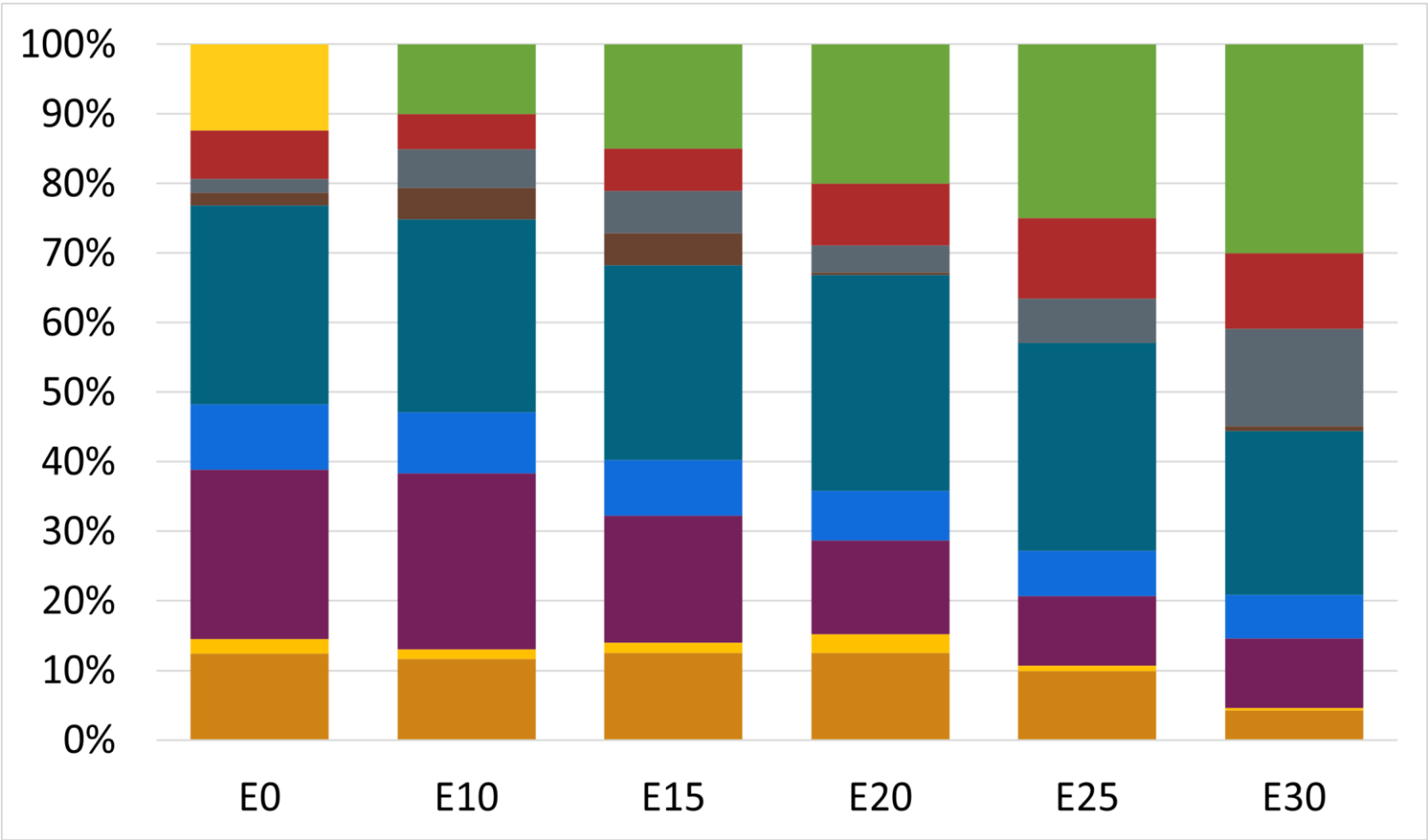
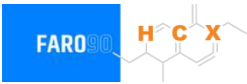


Regular Gasoline RP – Constant Octane



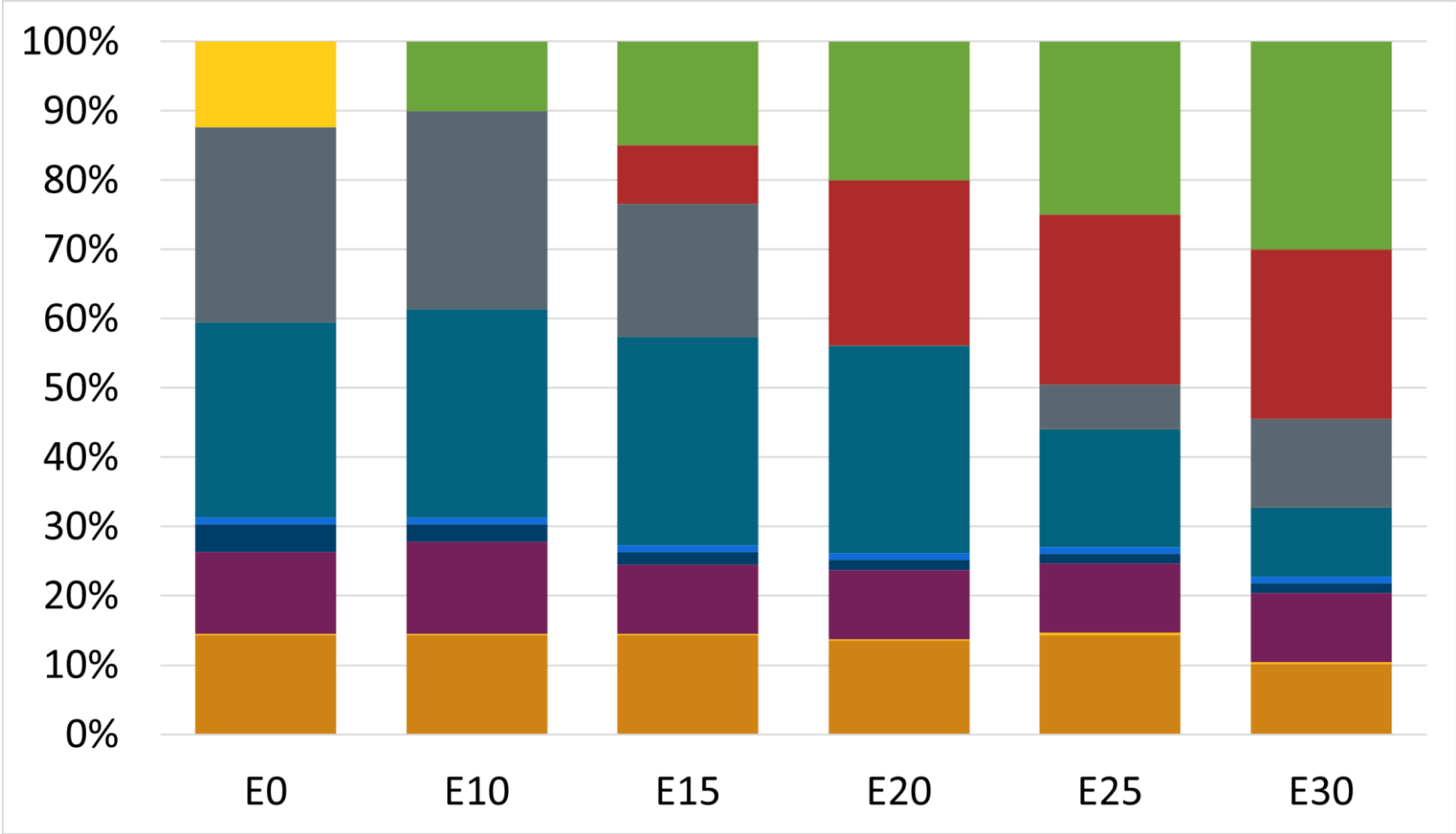
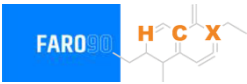
Octane (AKI)	87.1	87.0	87.0	87.0	87.0	87.0
Price (US\$/gal)	\$ 2.034	\$ 2.076	\$ 2.000	\$ 1.927	\$ 1.856	\$ 1.794

Premium Gasoline RP – Constant Octane



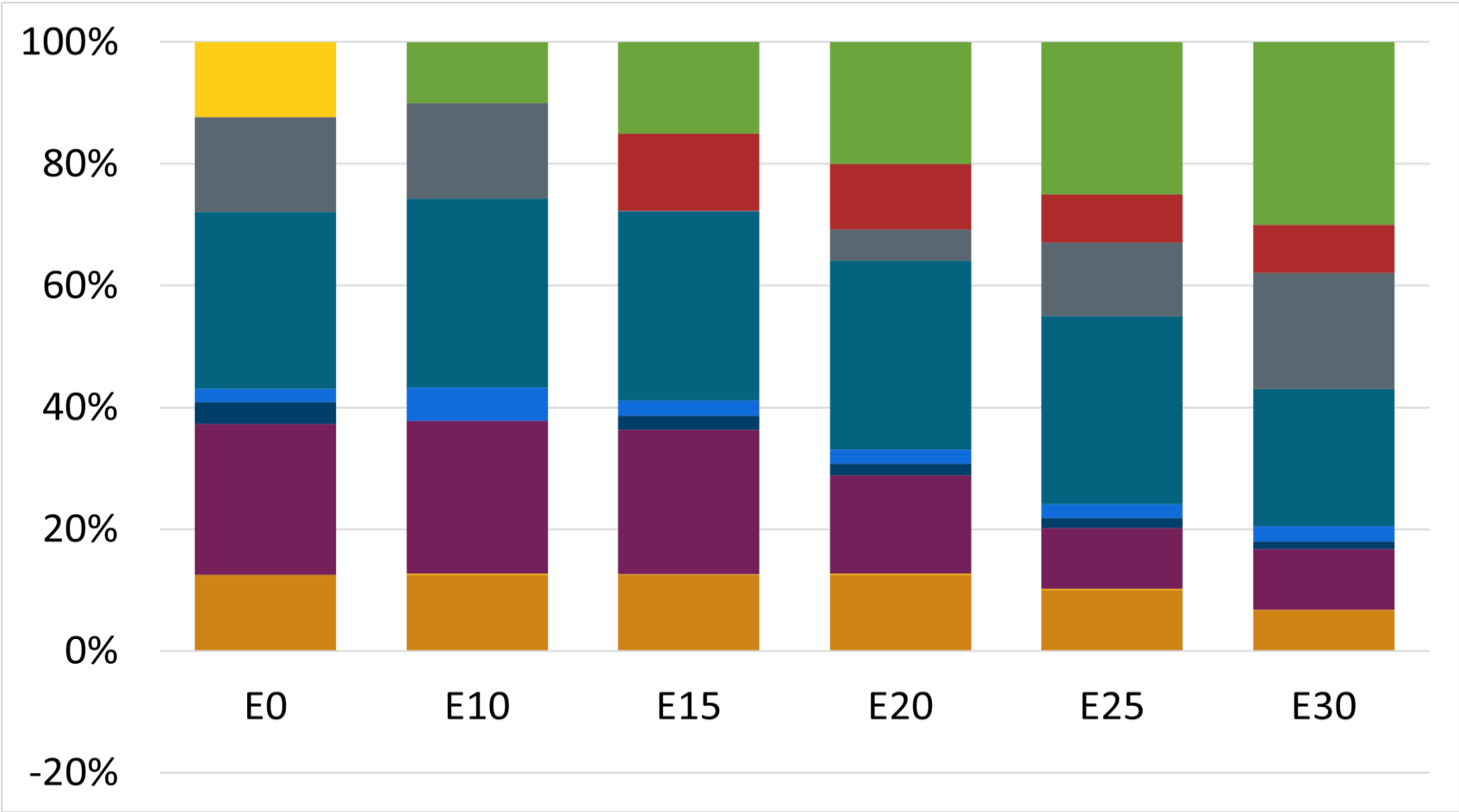
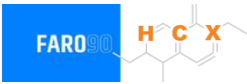
Octane (AKI)	91.0	91.0	91.0	91.0	91.0	91.0
Price (US\$/gal)	\$ 2.212	\$ 2.185	\$ 2.120	\$ 2.056	\$ 1.990	\$ 1.923

Regular Gasoline ZM – Constant Octane



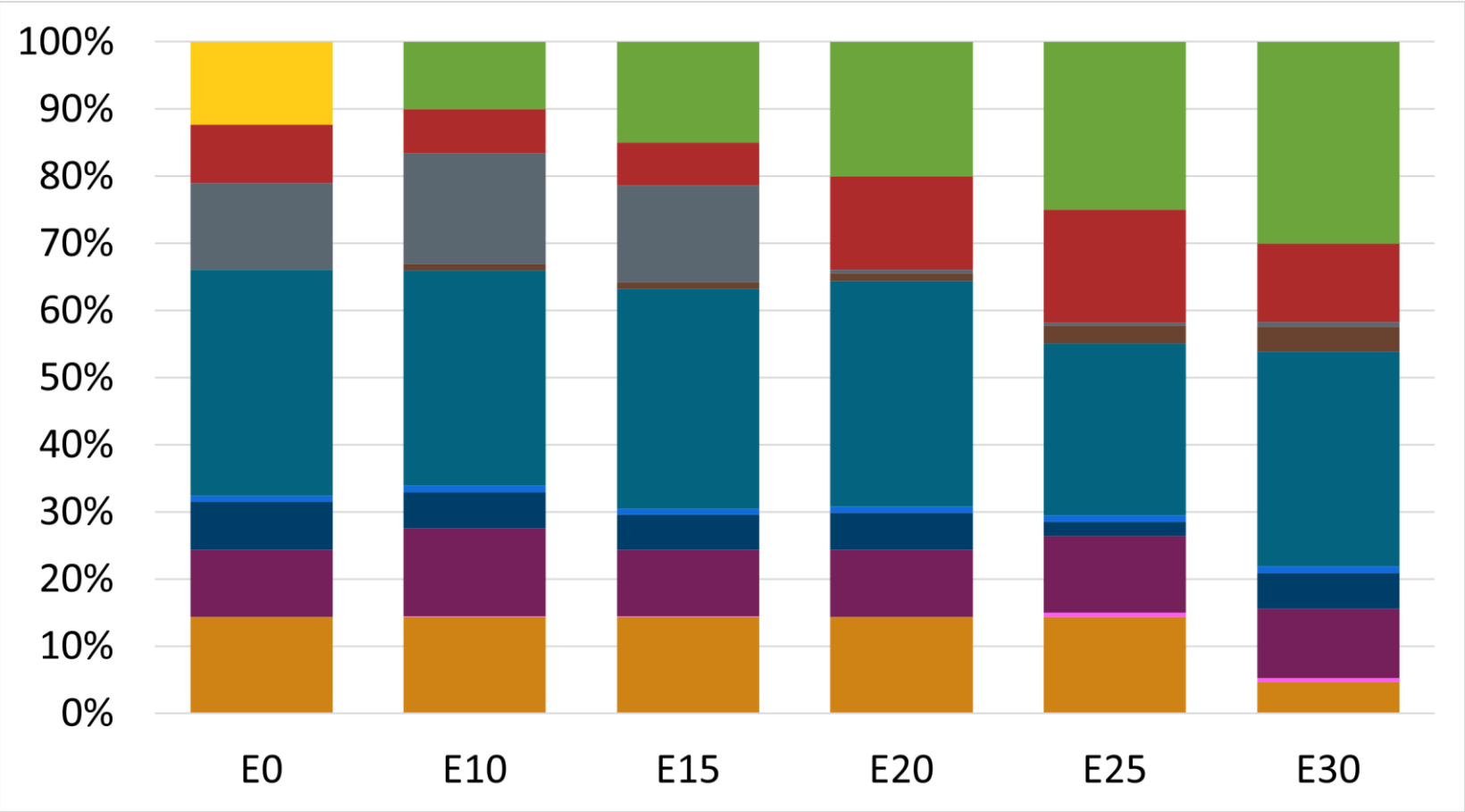
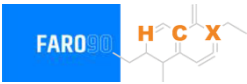
Octane (AKI)	87.0	87.0	87.0	87.0	87.0	87.0
Price (US\$/gal)	\$ 2.070	\$ 2.054	\$ 1.989	\$ 1.920	\$ 1.850	\$ 1.779

Premium Gasoline ZM – Constant Octane



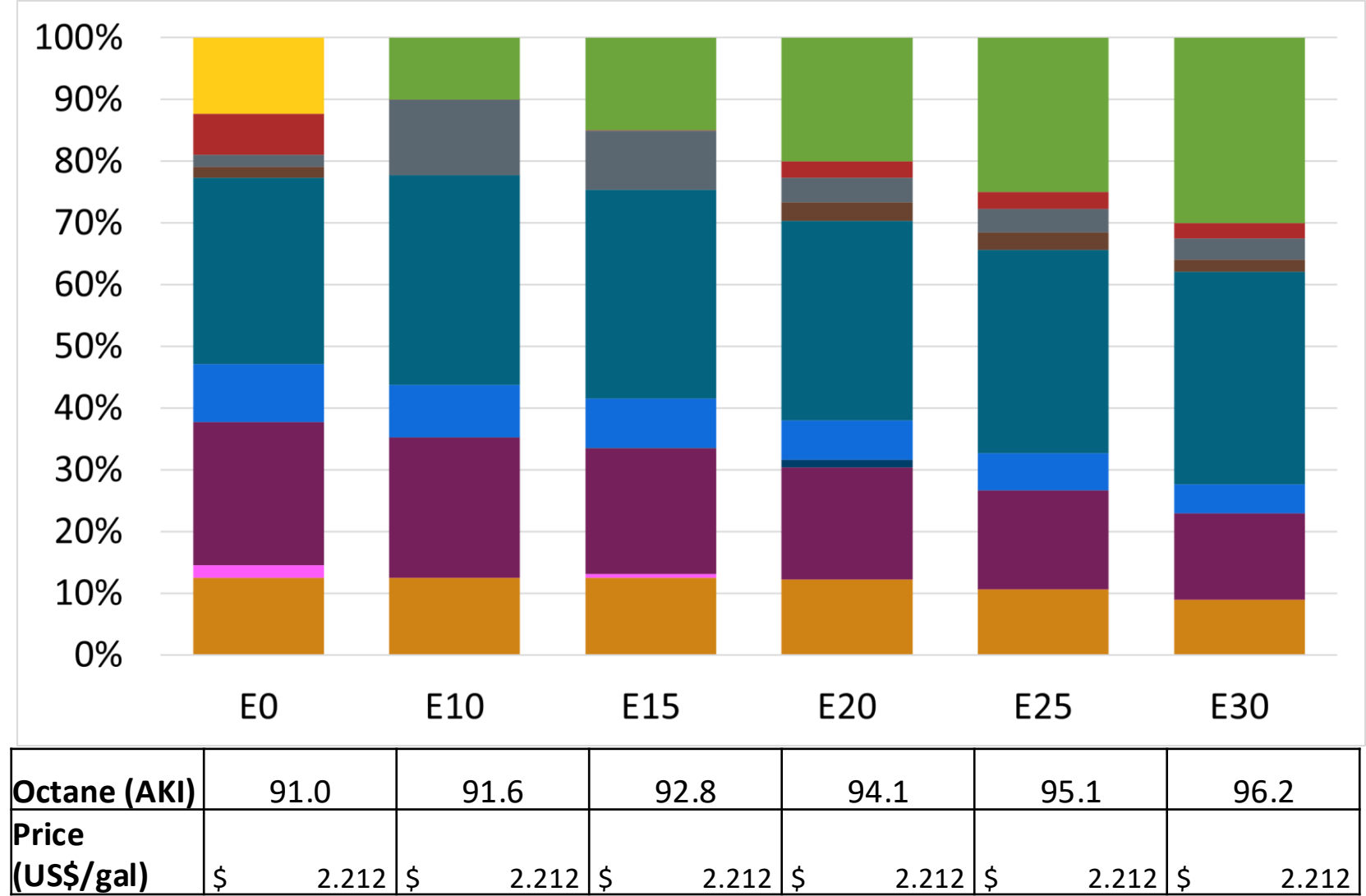
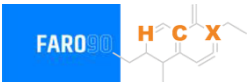
Octane (AKI)	91.0	91.0	91.0	91.0	91.0	91.0
Price (US\$/gal)	\$ 2.267	\$ 2.232	\$ 2.167	\$ 2.099	\$ 2.031	\$ 1.962

Regular Gasoline RP – Increased Octane

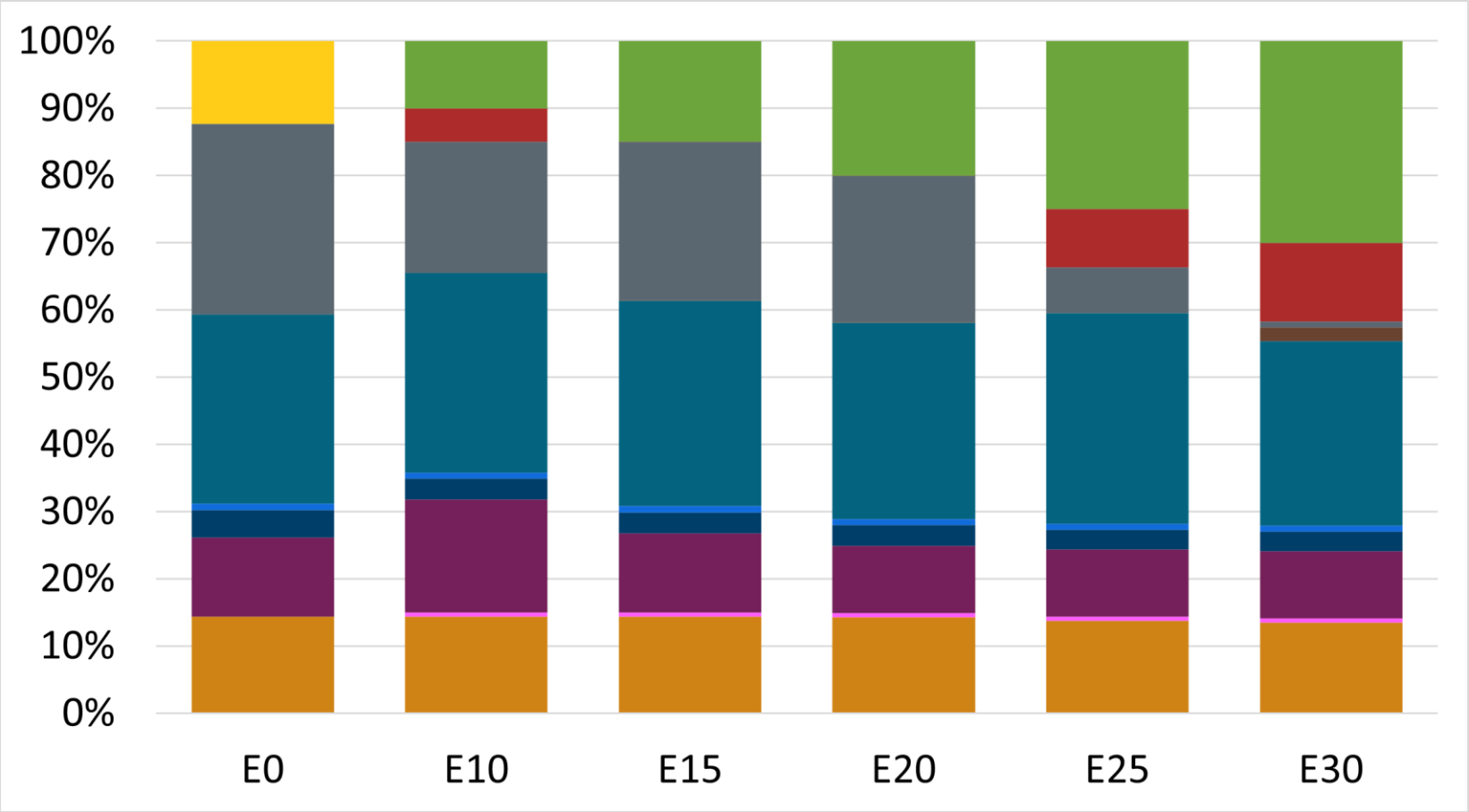


Octane (AKI)	87.0	87.2	88.6	90.1	90.5	92.9
Price (US\$/gal)	\$ 2.020	\$ 2.020	\$ 2.020	\$ 2.020	\$ 2.020	\$ 2.020

Premium Gasoline RP – Increased Octane

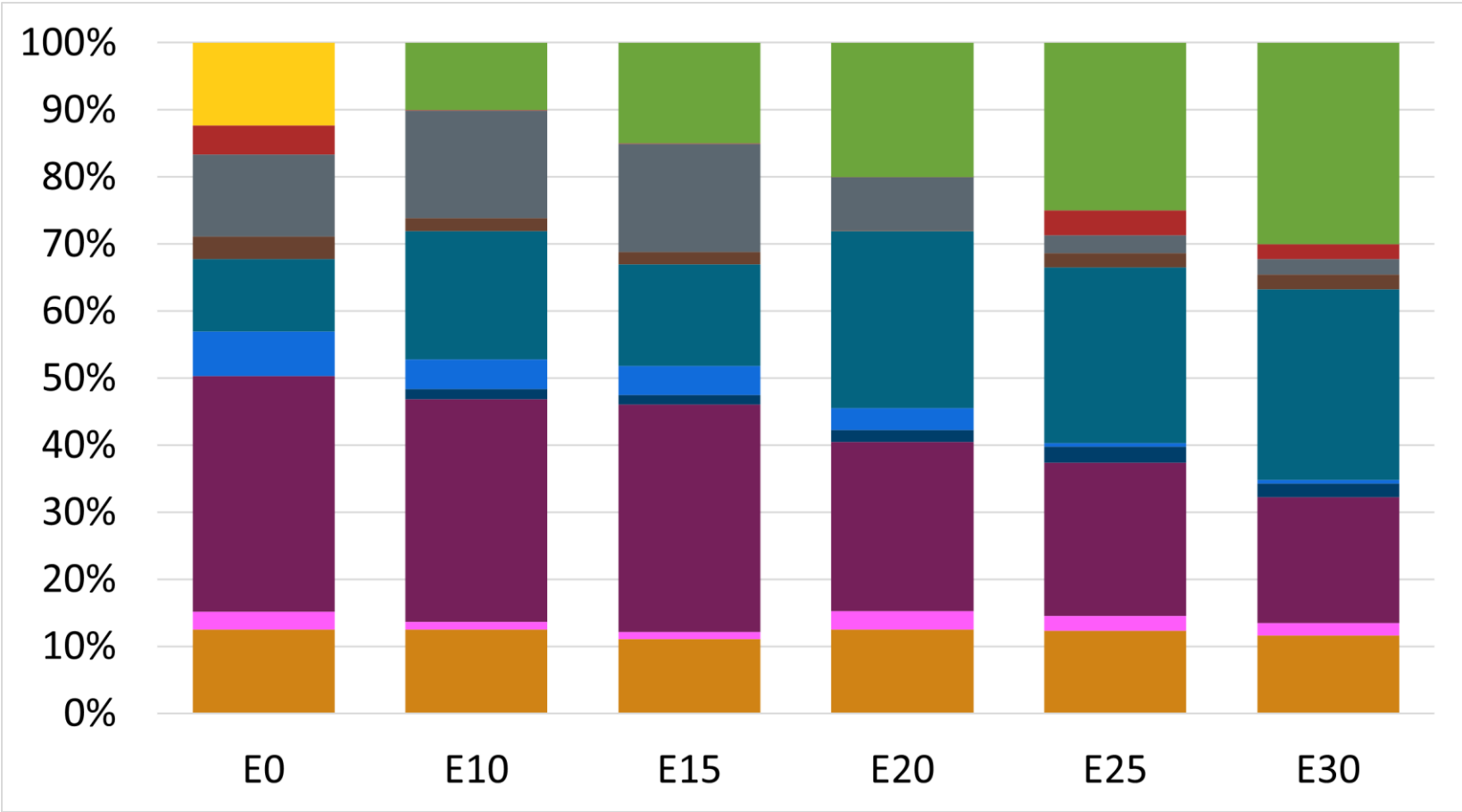
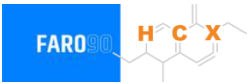


Regular Gasoline ZM – Increased Octane



Octane (AKI)	87.0	87.4	88.9	90.4	91.7	93.2
Price (US\$/gal)	\$ 2.070	\$ 2.070	\$ 2.070	\$ 2.070	\$ 2.070	\$ 2.070

Premium Gasoline ZM – Increased Octane



Octane (AKI)	91.0	91.6	93.0	94.3	95.1	96.5
Price (US\$/gal)	\$ 2.264	\$ 2.264	\$ 2.264	\$ 2.264	\$ 2.264	\$ 2.264

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

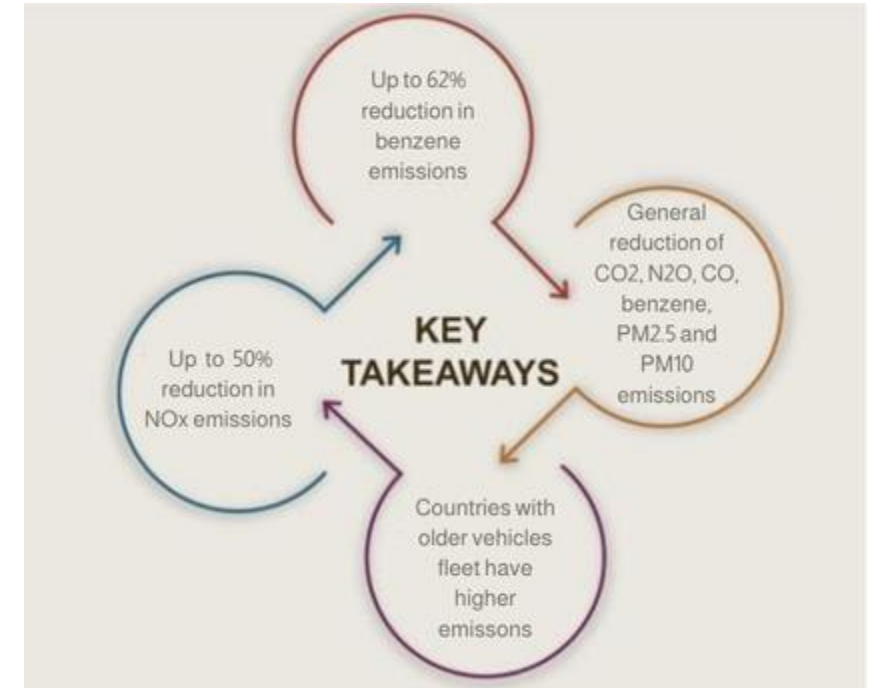
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Vehicular Fleet in Mexico

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles				
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	3,465				
	Single FI	3-Ways		> 8 years, > 160 tkm	89,129				
	Multi FI	3-Ways		> 8 years, > 160 tkm	3,808,157				
		Euro III		> 8 years, > 160 tkm	5,537,051				
		Euro IV		> 8 years, > 160 tkm	4,183,161				
		Euro V		> 8 years, > 160 tkm	5,781,437				
				4-8 years, 80-160 tkm	4,462,625				
		< 4 years, < 80 tkm		4,281,459					
	Hybrid			Any	234,946				
Electric			Any	32,663					
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	343				
	Single FI	3-Ways		> 8 years, > 160 tkm	2,799				
	Multi FI	3-Ways		> 8 years, > 160 tkm	664,855				
		Euro III		> 8 years, > 160 tkm	1,967,943				
		Euro IV		> 8 years, > 160 tkm	1,334,250				
		Euro V		> 8 years, > 160 tkm	2,074,483				
				4-8 years, 80-160 tkm	2,261,569				
		< 4 years, < 80 tkm		3,723,434					
	Hybrid			Any	175,050				
	Electric			Any	28,023				
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	53,390				
		Euro II		> 8 years, > 160 tkm	85,136				
		Euro III		> 8 years, > 160 tkm	1,896,819				
		Euro IV		> 8 years, > 160 tkm	1,024,779				
		Euro V		4-8 years, 80-160 tkm	1,287,524				
				4-8 years, 80-160 tkm	1,223,855				
				< 4 years, < 80 tkm	2,499,594				
	Electric			Any	1,614,420				
	Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	18,254			
FI		3-Ways	> 8 years, > 160 tkm		2,630,270				
		Euro III	> 8 years, > 160 tkm		2,338,059				
		Euro IV	> 8 years, > 160 tkm		1,551,482				
		Euro V / Euro 6	> 8 years, > 160 tkm		2,170,211				
			4-8 years, 80-160 tkm		1,601,860				
		< 4 years, < 80 tkm	2,814,954						
Hybrid			Any		5,074				
Electric			Any	1,015					

Gasoline Vehicular Fleet: **48,247,257**

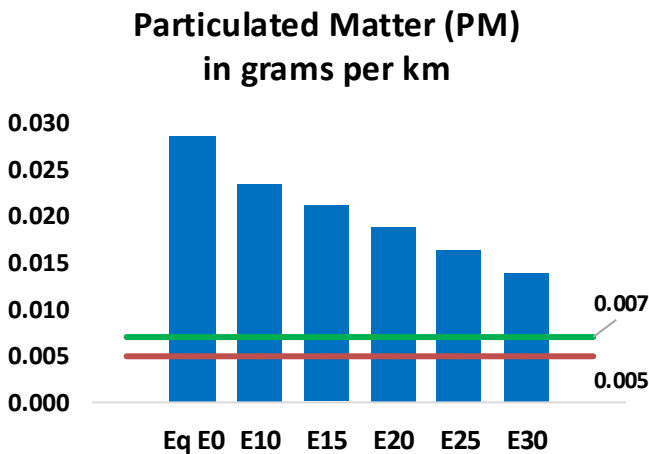
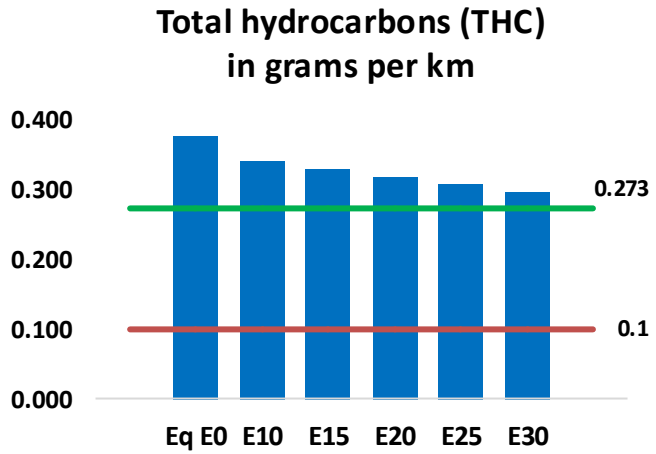
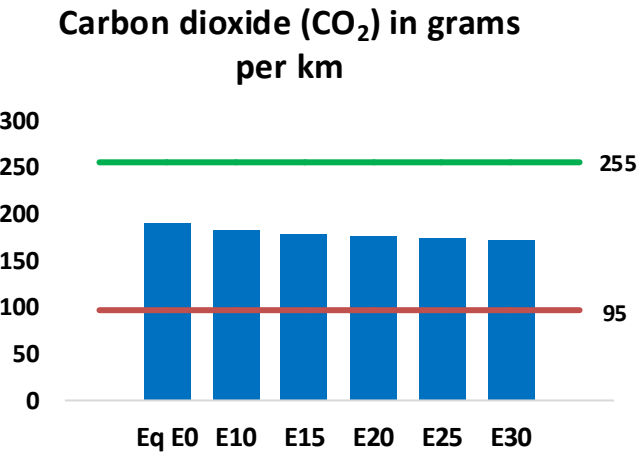
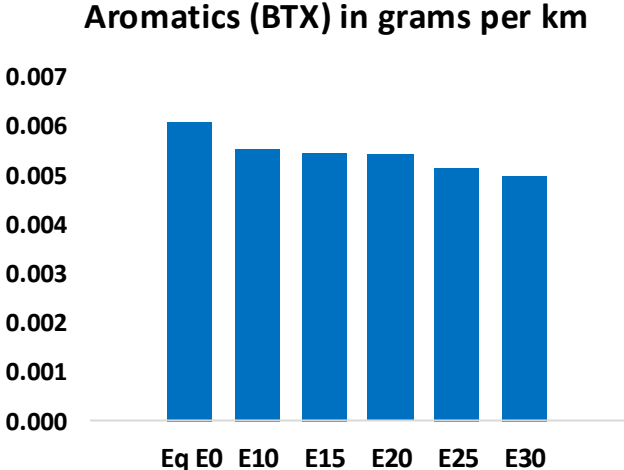
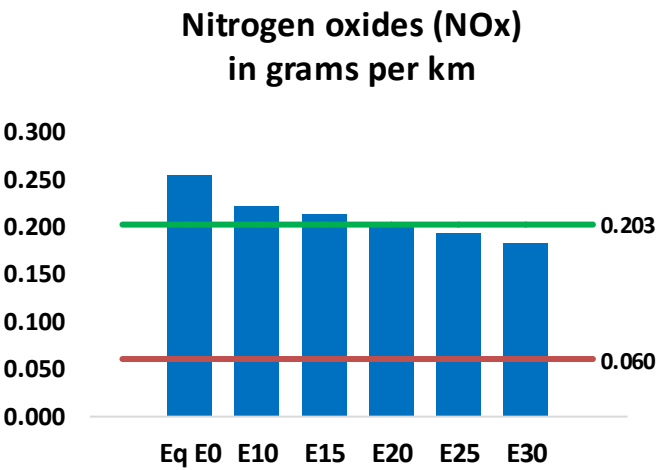
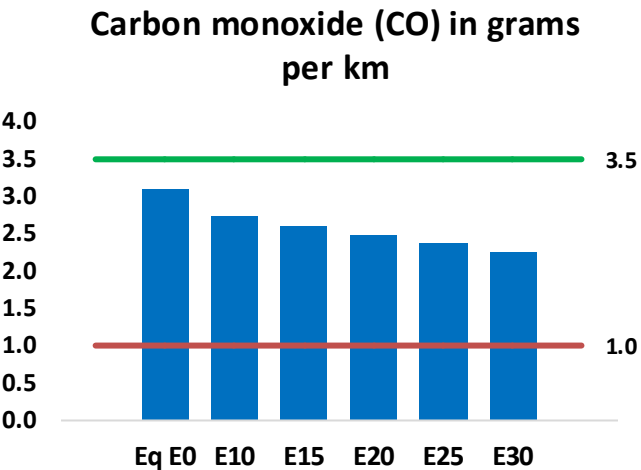
Average Age: **10 years**

Total = 63,463,539

Source: INEGI

Vehicular Emissions in Mexico

United States
Euro 6



Mexico – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	3,770,544	3,551,737	3,332,931	3,167,493	3,007,391	2,886,207	2,734,181	-12%	-20%
CO2	230,824,858	224,889,361	219,283,615	214,874,860	212,693,235	211,504,284	207,605,652	-5%	-8%
VOC	455,609	434,222	412,836	397,972	384,239	374,049	359,145	-9%	-16%
NOx	308,561	284,905	268,484	260,020	246,494	235,842	223,331	-13%	-20%
SOx	2,639	2,440	2,240	2,072	1,910	1,734	1,550	-15%	-28%
NH3	81,098	80,295	79,492	79,870	80,768	81,400	81,925	-2%	0%
N2O	3,065	3,049	3,037	3,051	3,114	3,149	3,185	-1%	2%
CH4	115,733	115,733	115,733	117,469	120,015	122,214	124,298	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	4,349	4,183	4,016	3,912	3,823	3,759	3,657	-8%	-12%
Acetaldehyde	10,181	11,685	17,146	26,111	35,573	40,649	48,032	68%	249%
Formaldehyde	35,972	34,973	40,769	47,871	50,152	55,482	60,398	13%	39%
Benzene	7,401	7,113	6,740	6,633	6,578	6,291	6,088	-9%	-11%
PM2.5	14,022	12,748	11,473	10,351	9,253	8,077	6,845	-18%	-34%
PM10	20,505	18,641	16,777	15,137	13,532	11,811	10,010	-18%	-34%

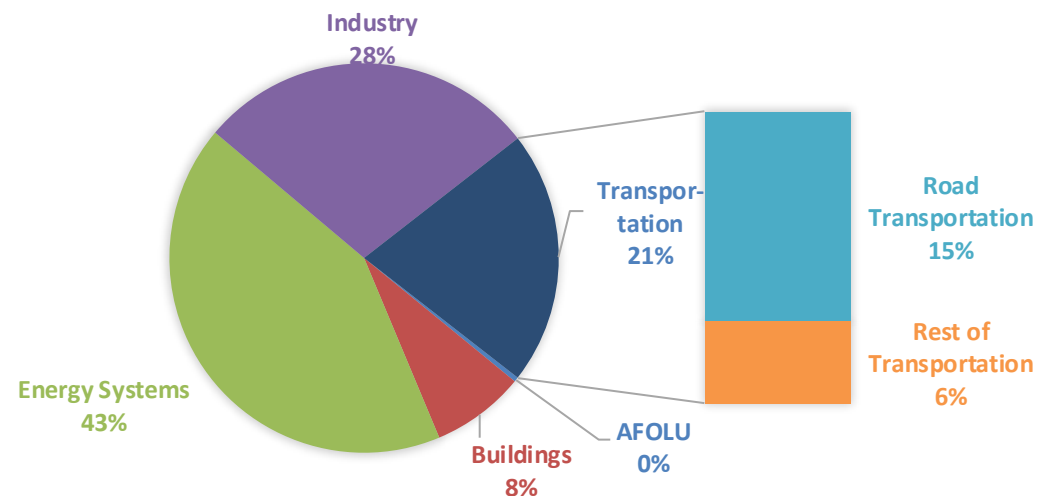
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

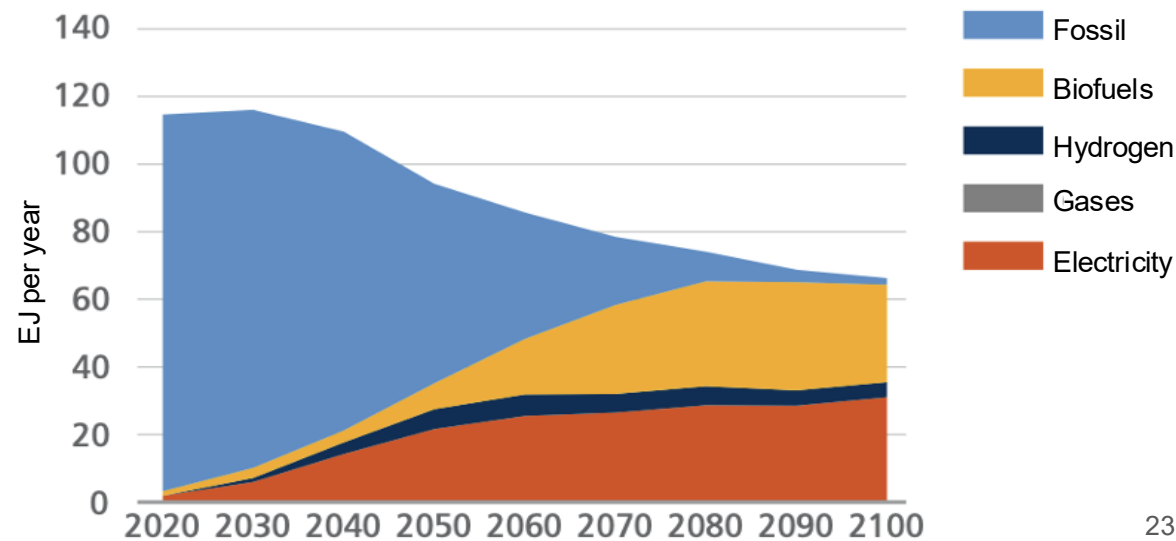
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

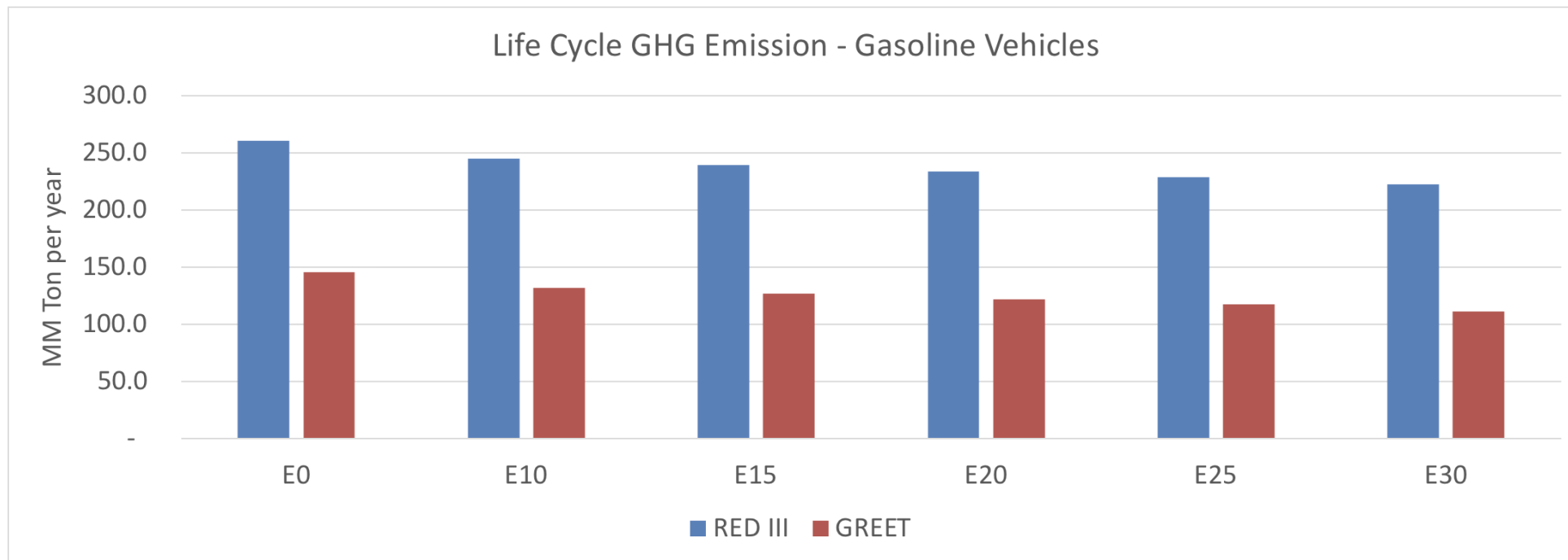
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Mexico – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	751.4
2035 Target (MMT/year)	500.4
Delta	251.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.4%	12.9%	16.4%	19.4%	23.4%
RED II/III	0.0%	5.9%	8.2%	10.3%	12.2%	14.6%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	5.4%	7.5%	9.5%	11.2%	13.5%
RED II/III	0.0%	6.1%	8.5%	10.7%	12.7%	15.2%